

LAB1: Understanding Devices & Ethernet Connection with RJ45

Objective

To study and understand the functioning and roles of the following network devices:

- Repeater
- Hub
- Switch
- Bridge
- Router
- Gateway

Apparatus (Software/Hardware)

- No software or hardware required for the theoretical understanding of network devices.

Part 1: Study of Network Devices

1. Repeater

- **Function:** Operates at the Physical Layer (Layer 1) of the OSI model.
- **Description:**
 - A repeater is an electronic device that amplifies and retransmits signals to extend their range.
 - It is used to overcome signal attenuation over long distances.
 - A repeater has two ports, so it can connect only two devices.

2. Hub

- **Function:** Operates at the Physical Layer (Layer 1) of the OSI model.
- **Description:**
 - A hub connects multiple Ethernet devices and makes them function as a single network segment.
 - It is essentially a multiport repeater.
 - Hubs detect collisions and forward a jam signal to all connected ports in case of a collision.

3. Switch

- **Function:** Operates at the Data Link Layer (Layer 2) and sometimes at the Network Layer (Layer 3) of the OSI model.
- **Description:**
 - A switch connects network segments and processes data packets to forward them to the correct destination.
 - Layer 2 switches operate as network bridges, while Layer 3 switches additionally handle routing.

4. Bridge

- **Function:** Operates at the Data Link Layer (Layer 2) of the OSI model.
- **Description:**
 - Bridges interconnect multiple network segments and filter data based on MAC addresses.
 - A bridge adheres to the IEEE 802.1D standard and is often considered synonymous with Layer 2 switches.

5. Router

- **Function:** Operates at the Network Layer (Layer 3) of the OSI model.
- **Description:**
 - A router interconnects two or more networks and forwards data packets based on their IP addresses.
 - Routers maintain routing tables and exchange information to determine optimal paths for data transmission.

6. Gateway

- **Function:** Operates across multiple layers depending on its implementation.
- **Description:**
 - A gateway connects networks using different protocols and performs protocol conversions for interoperability.
 - Gateways may include devices like protocol translators and signal converters.

Part 2: RJ45 Ethernet Connection

Objective

To study different types of network cables and practically implement the creation of cross-wired and straight-through cables using a crimping tool.

Apparatus (Components)

- RJ45 connectors
- Crimping tool
- Twisted pair cable

Procedure

1. Cable Preparation:

- Strip off about 2 inches of the plastic jacket at the end of the cable.
- Ensure not to nick or damage the internal wires during stripping.

2. Wire Arrangement:

- Spread the internal wires apart while holding the jacket to prevent further untwisting inside the cable.
- Maintain a maximum of ½ inch of untwisted wire to comply with standards.

3. Connector Installation:

- Determine the type of cable (straight-through or cross-wired).
- Arrange the wires in the appropriate sequence based on the chosen cable type (refer to standard wiring diagrams).

4. Crimping:

- Insert the arranged wires into the RJ45 connector.
- Use a crimping tool to secure the connector to the cable, ensuring proper alignment and contact.

5. Testing:

- Use a cable tester to verify the continuity and correctness of the wiring.

Wiring Diagram

Straight-Through Cable Wiring

Pin	Color	Pin	Color
1	White/Orange	1	White/Orange
2	Orange	2	Orange
3	White/Green	3	White/Green
4	Blue	4	Blue
5	White/Blue	5	White/Blue
6	Green	6	Green
7	White/Brown	7	White/Brown
8	Brown	8	Brown

Cross-Wired Cable Wiring

Pin	Color	Pin	Color
1	White/Orange	1	White/Green
2	Orange	2	Green
3	White/Green	3	White/Orange
4	Blue	4	Blue
5	White/Blue	5	White/Blue
6	Green	6	Orange
7	White/Brown	7	White/Brown
8	Brown	8	Brown

Diagram shows you how to prepare Cross wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Diagram shows you how to prepare straight through wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Conclusion

This lab provides an understanding of essential networking devices and the process of creating Ethernet cables using RJ45 connectors. By mastering these concepts, students gain foundational knowledge critical for computer networking.

LAB2: Networking Commands

In this lab, I explored and practiced essential networking commands that are commonly used to analyze, configure, and troubleshoot networks. I understood the purpose of each command and its output by working with real-world examples. Through this practical session, I gained confidence in interpreting network data and managing basic networking tasks efficiently.

Introduction:

Networking commands are essential tools that allow me to interact with and manage network configurations, monitor traffic, and troubleshoot issues effectively. They help in diagnosing connectivity problems, identifying the status of devices on a network, and ensuring that all configurations are working as expected. I understand that learning these commands is crucial for anyone working in IT or networking because they provide direct insight into how networks function and enable quick solutions to complex problems. These commands form the backbone of network administration and troubleshooting, making them a vital skill to master.

Practical Work:

1. ping Command

I used the ping command to test the connectivity between my device and another host. For example:

```
C:\Windows\System32\cmd.exe

D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>ping tufohss.edu.np

Pinging tufohss.edu.np [202.45.144.31] with 32 bytes of data:
Reply from 202.45.144.31: bytes=32 time=12ms TTL=53
Reply from 202.45.144.31: bytes=32 time=11ms TTL=53
Reply from 202.45.144.31: bytes=32 time=9ms TTL=53
Reply from 202.45.144.31: bytes=32 time=9ms TTL=53

Ping statistics for 202.45.144.31:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 12ms, Average = 10ms

D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>ping google.com

Pinging google.com [2404:6800:4002:821::200e] with 32 bytes of data:
Reply from 2404:6800:4002:821::200e: time=24ms
Reply from 2404:6800:4002:821::200e: time=20ms
Reply from 2404:6800:4002:821::200e: time=19ms
Reply from 2404:6800:4002:821::200e: time=26ms

Ping statistics for 2404:6800:4002:821::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 26ms, Average = 22ms
```

This command sends ICMP echo requests to the target IP address. I observed that if the host is reachable, it replies with echo responses, confirming the connection.

2. **ipconfig / ifconfig Command**

I used ipconfig to view my network configuration,

```
D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : worldlink.com.np
    IPv6 Address. . . . . : 2400:1a00:b080:5308::2
    IPv6 Address. . . . . : 2400:1a00:b080:5308:cc95:561:c9cb:ec92
    Temporary IPv6 Address. . . . . : 2400:1a00:b080:5308:75bc:89c0:1d20:62c9
    Link-local IPv6 Address . . . . . : fe80::33cb:e093:5c0b:b297%9
    IPv4 Address. . . . . : 192.168.1.84
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%9
                                192.168.1.254
```

These commands display my device's IP address, subnet mask, and default gateway. This helped me confirm my network setup and troubleshoot connectivity issues.

3. **tracert / traceroute Command**

I ran tracert to trace the route packets take to a destination, like:

```
D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>tracert youtube.com

Tracing route to youtube.com [2404:6800:4002:82b::200e]
over a maximum of 30 hops:

  1      2 ms      2 ms      1 ms     2400:1a00:b080:5308::1
  2      8 ms      4 ms      4 ms     2400:1a00:b1a7::1
  3      *         *         *        Request timed out.
  4     34 ms     25 ms     17 ms     2400:1a00:0:41::170
  5      8 ms     12 ms     17 ms     2400:1a00:0:41::128
  6      8 ms     11 ms     14 ms     2400:1a00:dccc:1:72:9:128:67
```

This command shows the path packets follow to reach the destination. It helped me understand the intermediate hops and where delays might occur.

4. nslookup Command

I used nslookup to find the IP address of a domain, for example:

```
D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>nslookup www.google.com
Server:  vip6-safenet-kmd01.wlink.com.np
Address:  2400:1a00:0:32::165

Non-authoritative answer:
Name:     www.google.com
Addresses: 2404:6800:4002:82e::2004
          142.250.206.100

D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>nslookup 142.250.206.100
Server:  vip6-safenet-kmd01.wlink.com.np
Address:  2400:1a00:0:32::165

Name:     del11s20-in-f4.1e100.net
Address:  142.250.206.100
```

This command queries the DNS server to resolve domain names into IP addresses. It's useful for troubleshooting DNS-related issues.

5. netstat Command

I explored active connections using netstat.

```
D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>netstat -an

Active Connections

    Proto Local Address           Foreign Address         State
    TCP    0.0.0.0:135             0.0.0.0:0               LISTENING
    TCP    0.0.0.0:445             0.0.0.0:0               LISTENING
    TCP    0.0.0.0:1521            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:5040            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:5357            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:7070            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:7680            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:8080            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:8090            0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49664           0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49665           0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49666           0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49667           0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49668           0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49679           0.0.0.0:0               LISTENING
    TCP    0.0.0.0:49682           0.0.0.0:0               LISTENING
    TCP    127.0.0.1:27017         0.0.0.0:0               LISTENING
    TCP    127.0.0.1:44950         0.0.0.0:0               LISTENING
    TCP    127.0.0.1:44960         0.0.0.0:0               LISTENING
    TCP    127.0.0.1:49670         0.0.0.0:0               LISTENING
```


This command shows open connections, listening ports, and routing tables. It gave me insight into the current network activity on my device.

6. **ipconfig /all**: Displays Address Resolution Protocol (ARP) path.

```
D:\5th Semester\Triton-College\ComputerNetworking\Arun Bhatt>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-2N91BA8
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : worldlink.com.np

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : B0-7B-25-89-23-21
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : D8-F8-83-E4-09-68
    DHCP Enabled. . . . . : Yes
```

By performing these commands, I developed a better understanding of how networks operate and how to analyze connectivity issues. This practical knowledge is foundational for any networking professional.

LAB3: Connecting Two LANs in Cisco Packet Tracer

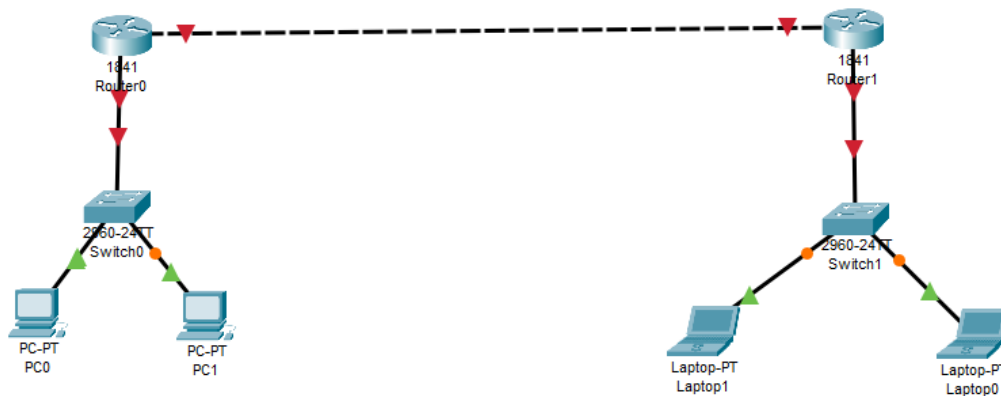
In this lab, I learned to connecting two separate Local Area Networks (LANs) using a router in Cisco Packet Tracer. I understand that LANs are essential for creating a network within a limited area, like an office or a home. Connecting two LANs allows devices in each network to communicate with each other, enabling resource sharing and data exchange. This setup demonstrates how routers manage communication between different subnets.

Practical Work

1. Setting Up the Topology

- **What I Did:**

- I created two LANs, each with a switch and three PCs.
- I connected the PCs to their respective switches using straight-through cables.
- For the router connection, I used cross-over cables to connect each switch to a router interface.



- **What I Learned:**

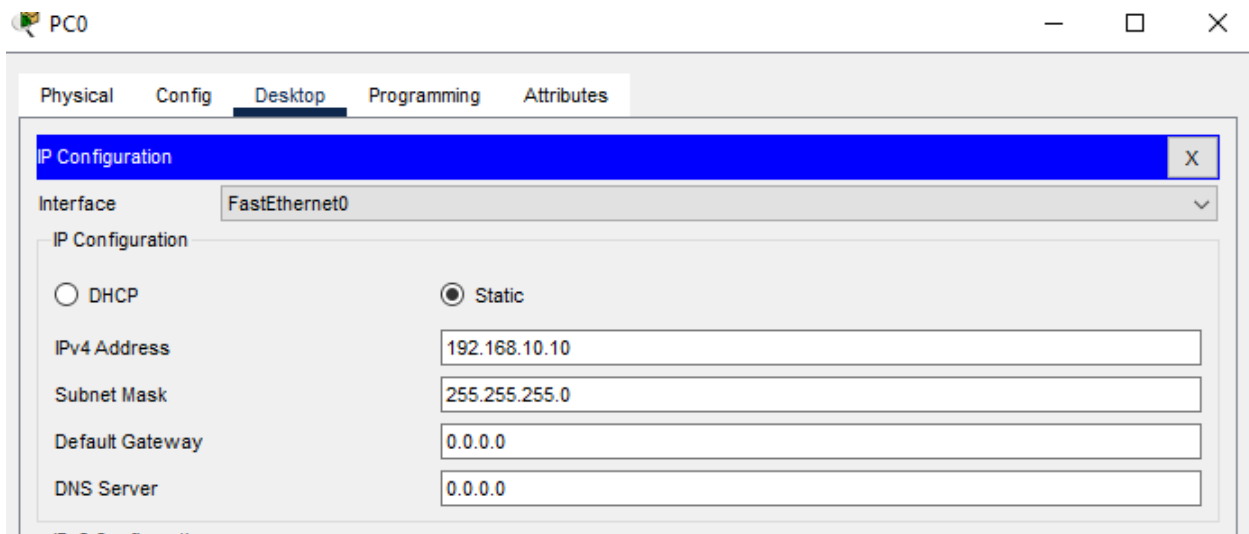
- Straight-through cables are used for connecting different types of devices (e.g., PC to switch), while cross-over cables connect similar devices (e.g., switch to router).
- A proper physical connection is the first step in establishing communication between networks.

2. Assigning IP Addresses

- **What I Did:**

- I assigned IP addresses to the PCs in each LAN using the **192.168.10.x** range for the first LAN and **192.168.20.x** range for the second LAN.

- Example configuration for PCs:
 - **LAN 1:**
 - PC1: 192.168.10.10
 - PC2: 192.168.10.11
 - **LAN 2:**
 - PC4: 192.168.20.10
 - PC5: 192.168.20.11



Same for others also

Each device in a LAN needs a unique IP address within the same subnet. The subnet mask (255.255.255.0) ensures all devices belong to the same network.

3. Configuring the Router

- **What I Did:**

- I accessed the router's CLI and configured the interfaces connected to each LAN with the respective subnet IP addresses.
- Commands I used:

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface FastEthernet0/0
```

```
Router(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
```

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.10.10 255.255.255.0
Router(config-if)#ip address 192.168.10.10 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

```
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.20.0 255.255.255.0
Bad mask /24 for address 192.168.20.0
Router(config-if)#ip address 192.168.20.10 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

Routers require an IP address for each connected network to act as a gateway for devices within those networks.

4. Testing Connectivity

- **What I Did:**

- I tested the connection by pinging a PC in LAN 2 from a PC in LAN 1. For example:

ping 192.168.20.10

- I also checked connectivity from LAN 2 to LAN 1 to confirm bidirectional communication.

```
C:\>ping 192.168.10.11

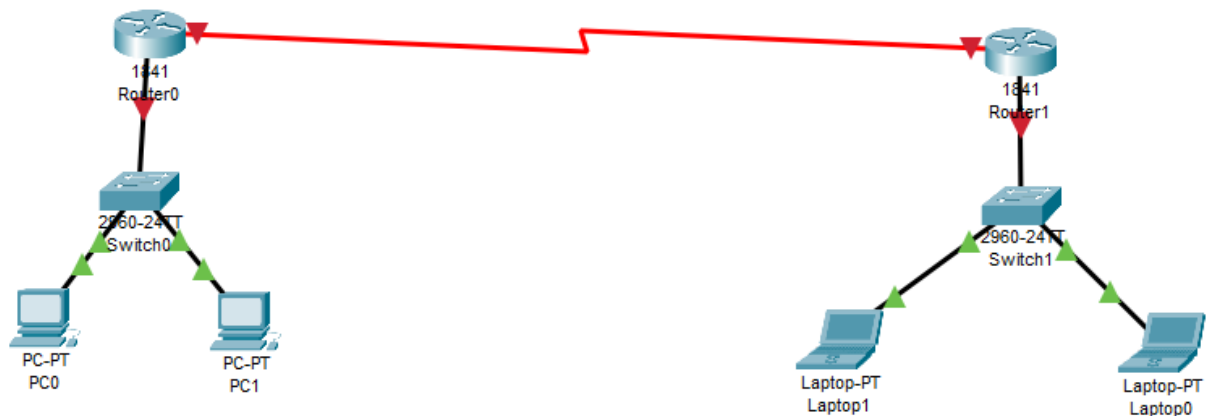
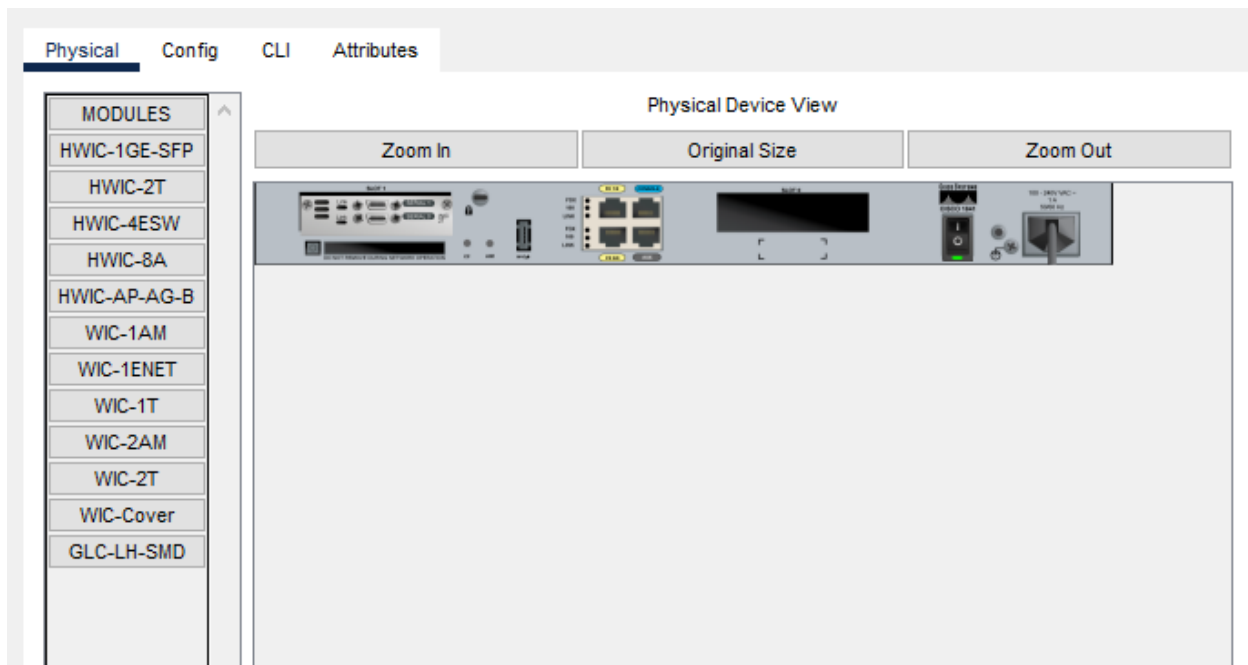
Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=5ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>|
```

No LAN setup:



```
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#ip address 192.168.10.10 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#int se 0/0
%Invalid interface type and number
Router(config)#int se 0/1/0
Router(config-if)#ip address 10.0.0.10 255.0.0.0
Router(config-if)#no shutdown
```

```

Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#ip address 192.168.20.10 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se 0/1/0
Router(config-if)#ip address 10.0.0.20 255.0.0.0
Router(config-if)#no shutdown

```

Physical **Config** CLI Attributes

GLOBAL ^

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

Static Routes

Network 192.168.10.0

Mask 255.255.255.0

Next Hop 10.0.0.20

Add

Physical **Config** CLI Attributes

GLOBAL ^

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

Static Routes

Network 192.168.20.0

Mask 255.255.255.0

Next Hop 10.0.0.10

Add

```

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.10.10: bytes=32 time=8ms TTL=255
Reply from 192.168.10.10: bytes=32 time=4ms TTL=255

```

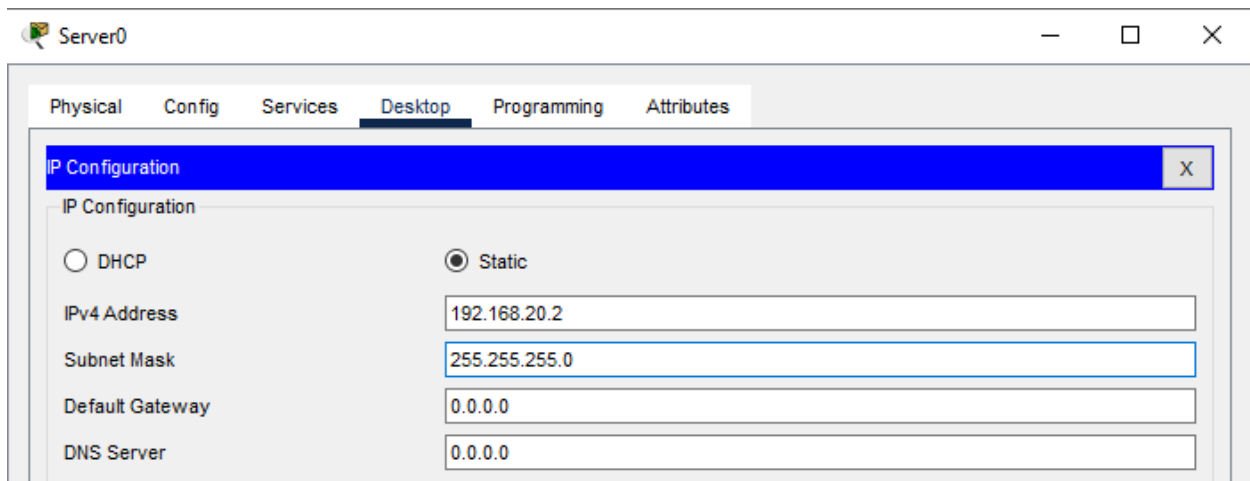
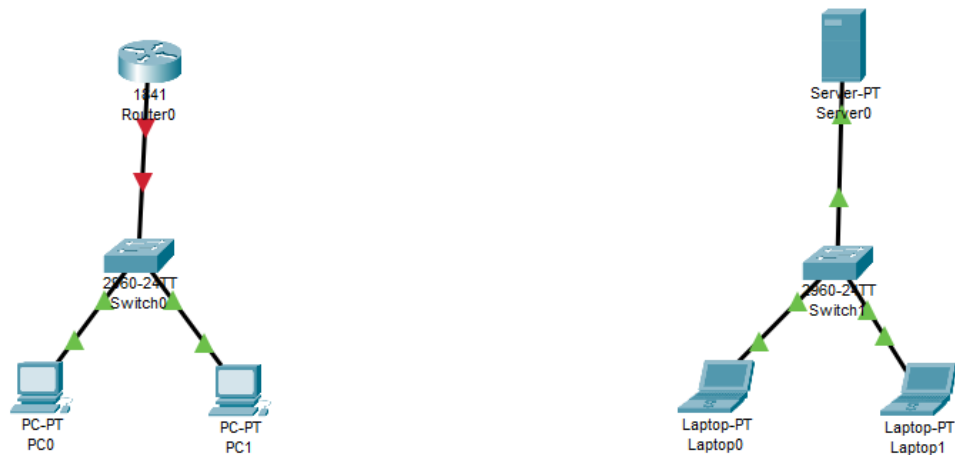
LAB3: LAN Configuration Using DHCP Protocol

In this lab, I configured a LAN using the Dynamic Host Configuration Protocol (DHCP) in Cisco Packet Tracer. DHCP simplifies the process of assigning IP addresses to devices within a network by automating it. This eliminates the need for manual configuration and ensures that no IP conflicts occur in the network.

Practical Work:

1. Setting Up the Topology

- Place a router, a switch, and two PCs in the workspace.
- Connect the PCs to the switch using straight-through cables.
- Connect the switch to the router using a straight-through cable.



Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.20.1

DNS Server: 192.168.20.2

Start IP Address: 169 254 0 0

Subnet Mask: 255 255 0 0

Maximum Number of Users: 512

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	169.254....	255.255....	512	0.0.0.0	0.0.0.0

- Define the DHCP pool and assign the subnet details:

Laptop0

Physical **Config** Desktop Programming Attributes

GLOBAL

- Settings**
- Algorithm Settings

INTERFACE

- FastEthernet0
- Bluetooth

Global Settings

Display Name: Laptop0

Interfaces: FastEthernet0

Gateway/DNS IPv4

☒ **DHCP**

☐ Static

Default Gateway: 0.0.0.0

DNS Server:

Router(config)# ip dhcp pool LAN1


```
Router(dhcp-config)# network 192.168.10.1 255.255.255.0
```

```
Router(dhcp-config)# default-router 192.168.10.1
```

```
Router(dhcp-config)# dns-server 8.8.8.8
```

- Exclude addresses to reserve them for static assignments, such as for the router itself:

```
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.5
```

- Assign the IP address to the router interface:

```
Router(config)# interface FastEthernet0/0
```

```
Router(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int fa 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
|
```

3. Configuring the PCs to Use DHCP

- On each PC, go to the **Desktop** tab and select **IP Configuration**.
- Set the IP Configuration to **DHCP**.

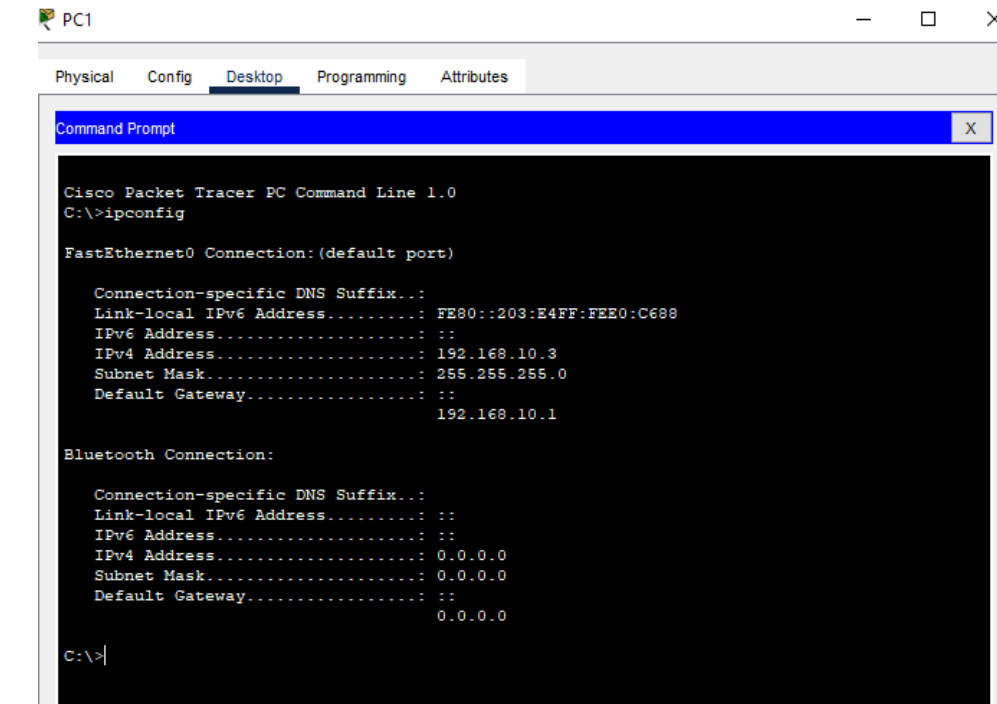
```
Router(config)#
Router(config)#Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool arun_pool
Router(dhcp-config)#network 192.168.10.1
% Incomplete command.
Router(dhcp-config)#network 192.168.10.1 255.255.255.0
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#defaultl-router 192.168.10.1
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router 192.168.10.1
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.10.5
Router(dhcp-config)#Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.2 255.255.255.0
Router(config-if)#Enter configuration commands, one per line. End with CNTL/Z.
Router(config-if)#
```

4. Testing the Configuration

- Verify that each PC has received an IP address automatically from the DHCP server by checking the assigned IP in the IP Configuration section of each PC.
- Test connectivity between the PCs using the ping command:



The screenshot shows the 'PC1' window in Cisco Packet Tracer. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the output of the 'ipconfig' command, showing the configuration for the 'FastEthernet0' connection. The IPv4 address is 192.168.10.3, and the default gateway is 192.168.10.1. The Bluetooth connection is also shown with all-zero values.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

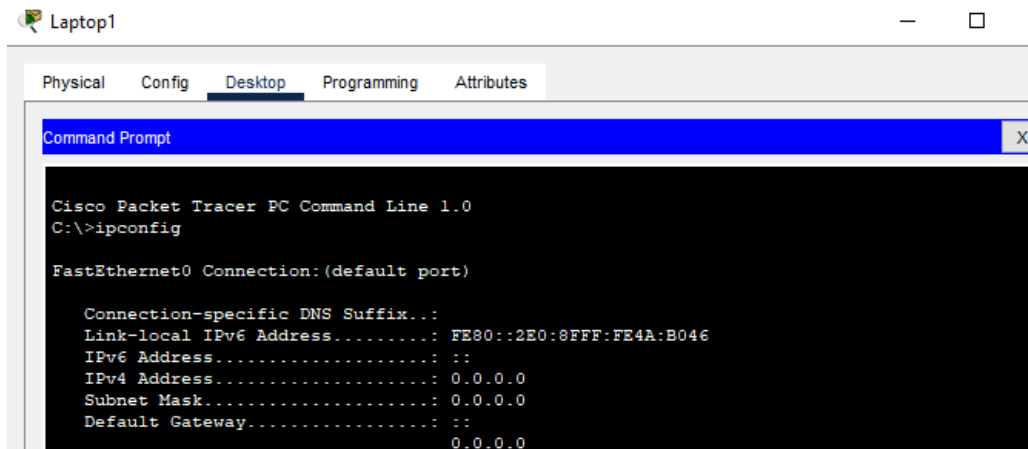
FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::203:E4FF:FEE0:C688
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```



The screenshot shows the 'Laptop1' window in Cisco Packet Tracer. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the output of the 'ipconfig' command, showing the configuration for the 'FastEthernet0' connection. The IPv4 address is 0.0.0.0, and the default gateway is 0.0.0.0. The Bluetooth connection is also shown with all-zero values.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:8FFF:FE4A:B046
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

Conclusion

This lab demonstrates the use of DHCP to automate IP address assignment in a LAN. It highlights how DHCP eliminates manual errors and simplifies network management, especially in larger networks.

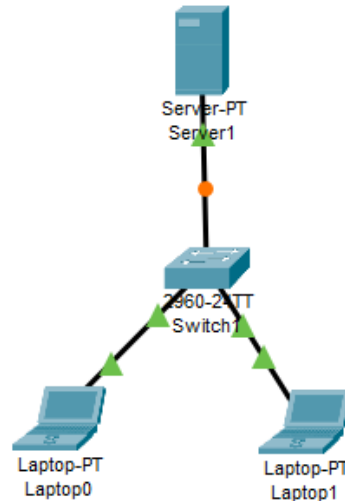
LAB5: Configuring DNS in a LAN Network

In this lab, I set up and configured a DNS (Domain Name System) server within a LAN network using Cisco Packet Tracer. DNS translates human-readable domain names into IP addresses, simplifying network navigation for users. Configuring DNS ensures that devices in the LAN can resolve domain names to their corresponding IP addresses effectively.

Practical Work

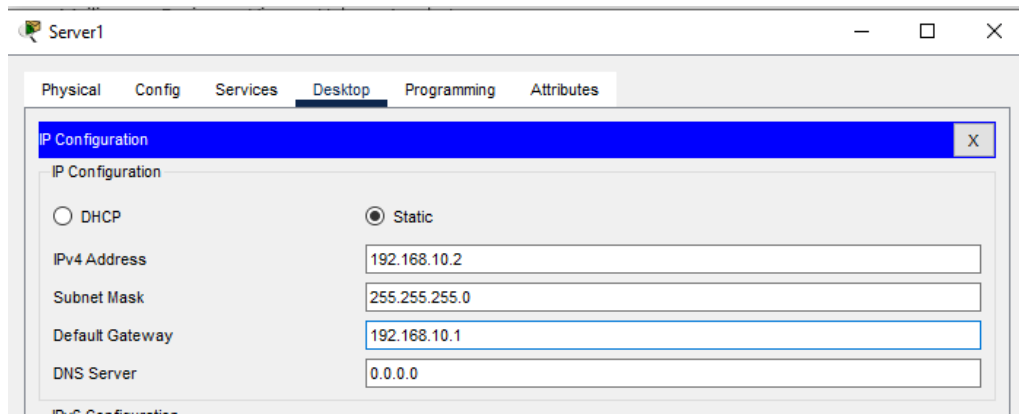
1. Setting Up the Topology

- Place a router, a switch, a server, and two PCs in the workspace.
- Connect the PCs and the server to the switch using straight-through cables.
- Connect the switch to the router using a straight-through cable.



2. Assigning Static IP Addresses

- Assign a static IP address to the DNS server:
 - IP Address: 192.168.10.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1



1. Configuring the DNS Server

- On the server, go to the **Services** tab and select **DNS**.
- Turn on the DNS service by toggling the **On/Off** switch.
- Add DNS records for domain name resolution:
 - For example, add www.example.com pointing to 192.168.10.10 (PC1's IP address).
 - Add more records as needed for other devices in the network.

The screenshot shows the 'Server1' configuration window with the 'Services' tab selected. In the left sidebar, 'DNS' is highlighted under the 'SERVICES' section. The main area shows the 'DNS' configuration. The 'DNS Service' is turned 'On'. Under 'Resource Records', a record is added with 'Name' 'arunbhatt', 'Type' 'A Record', and 'Address' '192.168.10.2'. Below this, there are 'Add', 'Save', and 'Remove' buttons. At the bottom, a table lists the configured records.

No.	Name	Type	Detail
0	arunbhatt	A Record	192.168.10.2

2. Configuring the PCs

- On each PC, go to the **Desktop** tab and select **IP Configuration**.
- Assign IP addresses dynamically using DHCP or manually configure them.
- Set the DNS server IP address as 192.168.10.2.

3. Testing the Configuration

- Open the command prompt on a PC.
- Test domain name resolution using the ping command:

```

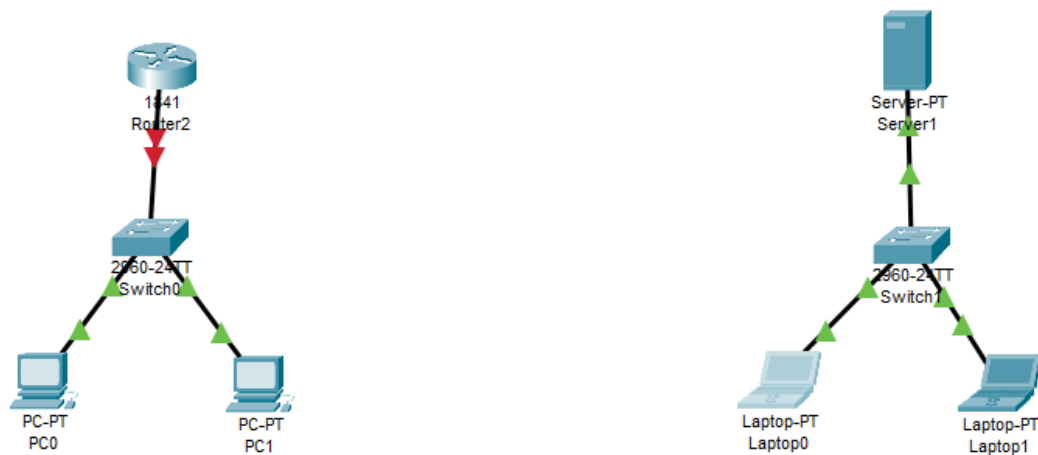
C:\>ping arunbhatt
Ping request could not find host arunbhatt. Please check the name and try again.
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128

```

Same configuration for Router also:



- Assign the router an IP address to act as the gateway:
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown

Configuring the DNS Server

- On the server, go to the **Services** tab and select **DNS**.
- Turn on the DNS service by toggling the **On/Off** switch.
- Add DNS records for domain name resolution:
 - For example, add www.example.com pointing to 192.168.10.10 (PC1's IP address).
 - Add more records as needed for other devices in the network.

Configuring the PCs

- On each PC, go to the **Desktop** tab and select **IP Configuration**.

- Assign IP addresses dynamically using DHCP or manually configure them.
- Set the DNS server IP address as 192.168.10.2.

Testing the Configuration

- Open the command prompt on a PC.
- Test domain name resolution using the ping command:

>> ping arunbhatt.local

- If configured correctly, the command should resolve the domain name to its IP address and successfully send packets to the target device.

If configured correctly, the command should resolve the domain name to its IP address and successfully send packets to the target device.

Conclusion

This lab demonstrates how to configure and use a DNS server in a LAN network. DNS simplifies communication by resolving domain names into IP addresses, enabling efficient network navigation and resource access.

LAB6: Configuring RIP Protocol in a Network

In this lab, I implemented the Routing Information Protocol (RIP) in a network using Cisco Packet Tracer. RIP is a dynamic routing protocol used to enable communication between routers in a network by exchanging routing tables. It operates based on hop count and is ideal for small to medium-sized networks. Configuring RIP ensures efficient routing without manual intervention for every new route.

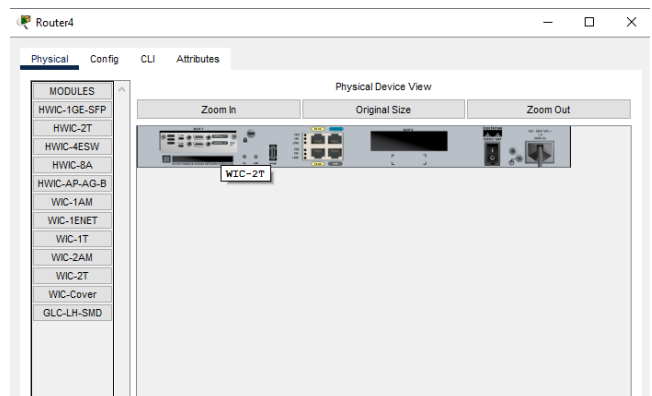
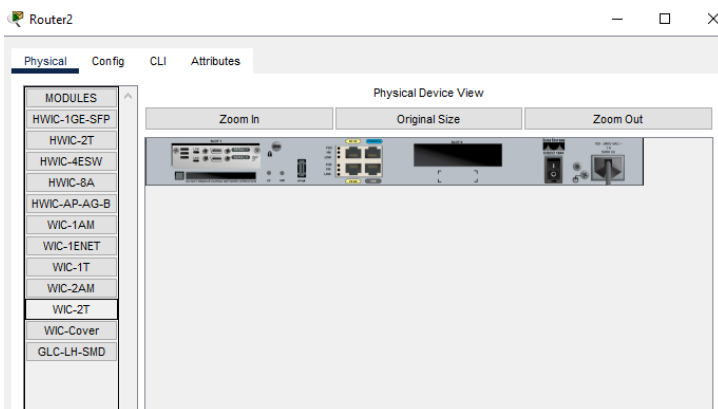
Practical Work

1. Setting Up the Topology

- I created a topology with three routers connected in a triangular configuration.
- Connected each router to its respective network (LAN) using switches and PCs.
- Used straight-through cables to connect PCs to switches and serial cables to interconnect routers.

2. Assigning IP Addresses

- Configured IP addresses for the routers' interfaces and PCs.
- Example IP scheme:
 - **Router 1:**
 - LAN: 192.168.10.0/24
 - Serial link to Router 2: 192.168.20.0/30
 - Serial link to Router 3: 192.168.30.0/30
 - **Router 2:**
 - LAN: 192.168.40.0/24
 - Serial link to Router 1: 192.168.20.0/30
 - Serial link to Router 3: 192.168.50.0/30



3. Configuring the Routers

- Accessed the CLI for each router and configured RIP:

```
Router> enable
```

```
Router# configure terminal
```

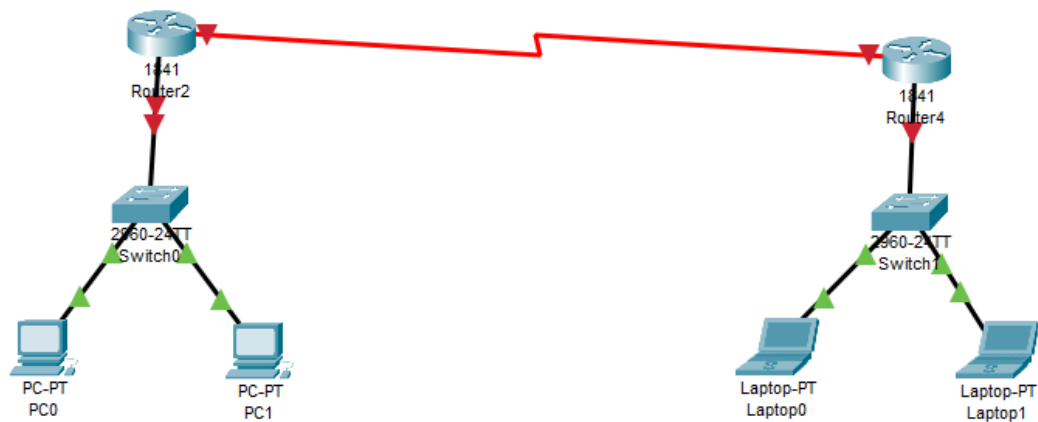
```
Router(config)# router rip
```

```
Router(config-router)# version 2
```

```
Router(config-router)# network 192.168.10.0
```

```
Router(config-router)# network 192.168.20.0
```

```
Router(config-router)# network 192.168.30.0
```



- Repeated the process for all routers, adding the respective network addresses connected to each router.

4. Enabling Interfaces

- Enabled the serial and FastEthernet interfaces on each router:

```
Router(config)# interface Serial0/0
```

```
Router(config-if)# ip address 192.168.20.1 255.255.255.252
```

```
Router(config-if)# no shutdown
```

5. Verifying RIP Configuration

- Checked the routing table on each router to ensure RIP learned the routes dynamically:

```
Router# show ip route
```


- Verified connectivity between PCs in different networks using the ping command:
ping 192.168.60.10

```
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.10.11
% Incomplete command.
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se 0/1/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#
```

Copy

Paste

```
C:\>ping 192.168.60.10

Pinging 192.168.60.10 with 32 bytes of data:

Reply from 192.168.60.10: bytes=32 time<1ms TTL=128
Reply from 192.168.60.10: bytes=32 time<1ms TTL=128
Reply from 192.168.60.10: bytes=32 time<1ms TTL=128
Reply from 192.168.60.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.60.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Conclusion

By configuring RIP in this lab, I enabled dynamic routing between multiple networks. The protocol ensured that routers exchanged routing information and determined the best path to each destination based on the number of hops. This approach simplifies routing in small to medium-sized networks.

LAB7: Configuring OSPF Protocol in a Network

Introduction

In this lab, I configured the Open Shortest Path First (OSPF) protocol to establish communication between two networks. OSPF is a dynamic routing protocol that uses a link-state routing algorithm, making it more efficient and scalable compared to RIP. It calculates the shortest path to a destination based on the cost of links and is widely used in larger and more complex networks.

Lab 6: Configuring OSPF Protocol in a Network

Introduction

In this lab, I configured the Open Shortest Path First (OSPF) protocol to establish communication between two networks. OSPF is a dynamic routing protocol that uses a link-state routing algorithm, making it more efficient and scalable compared to RIP. It calculates the shortest path to a destination based on the cost of links and is widely used in larger and more complex networks.

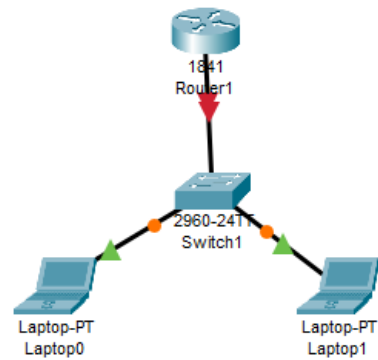
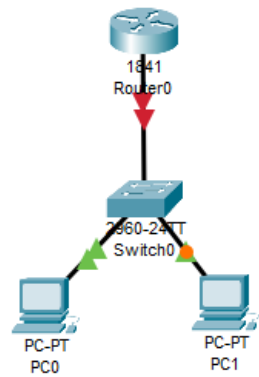
Network Topology

- **Network 1:**
 - 2 PCs connected to a switch, which is connected to a router (Router 1, model 1841).
 - IP range: **192.168.10.0/24**.
- **Network 2:**
 - 2 laptops connected to another switch, which is connected to a router (Router 2, model 1841).
 - IP range: **192.168.20.0/24**.

Practical Work

1. Physical Setup

- Connected PCs to the switch in Network 1 using straight-through cables.
- Connected the switch to Router 1 (FastEthernet0/0).
- Repeated the setup for Network 2 using laptops, a switch, and Router 2.
- Connected Router 1 to Router 2 using a serial cable (Serial0/0).



2. Assigning IP Addresses

- **Router 1 Configuration:**

- Accessed the CLI of Router 1 and assigned IP addresses to the interfaces

```
Router1> enable
```

```
Router1# configure terminal
```

```
Router1(config)# interface FastEthernet0/0
```

```
Router1(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Router1(config-if)# no shutdown
```

```
Router1(config)# interface Serial0/0
```

```
Router1(config-if)# ip address 192.168.30.1 255.255.255.252
```

```
Router1(config-if)# clock rate 64000
```

```
Router1(config-if)# no shutdown
```

- **Router 2 Configuration:**

- Assigned IP addresses to Router 2 interfaces

```
Router2> enable
```

```
Router2# configure terminal
```

```
Router2(config)# interface FastEthernet0/0
```

```
Router2(config-if)# ip address 192.168.20.1 255.255.255.0
```

```
Router2(config-if)# no shutdown
```

```
Router2(config)# interface Serial0/0
Router2(config-if)# ip address 192.168.30.2 255.255.255.252
Router2(config-if)# no shutdown
```

- **PCs and Laptops Configuration:**

- Set static IP addresses for PCs and laptops:
 - PC1: 192.168.10.10, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1
 - PC2: 192.168.10.11, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1
 - Laptop1: 192.168.20.10, Subnet Mask: 255.255.255.0, Gateway: 192.168.20.1
 - Laptop2: 192.168.20.11, Subnet Mask: 255.255.255.0, Gateway: 192.168.20.1

3. Configuring OSPF on Routers

- **Router 1 OSPF Configuration:**

- Configured OSPF with process ID 1:

```
Router1(config)# router ospf 1
Router1(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router1(config-router)# network 192.168.30.0 0.0.0.3 area 0
```

- **Router 2 OSPF Configuration:**

- Configured OSPF with the same process ID and areas:

```
Router2(config)# router ospf 1
Router2(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router2(config-router)# network 192.168.30.0 0.0.0.3 area 0
```

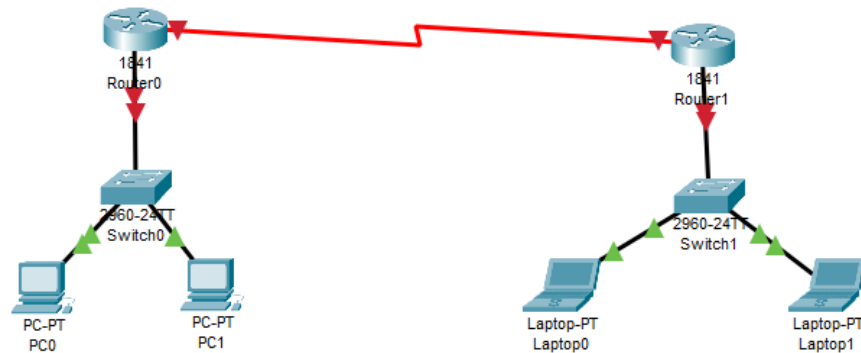
4. Verifying OSPF Configuration

- Checked OSPF neighbor relationships:

```
Router1# show ip ospf neighbor
Router2# show ip ospf neighbor
```

Testing Connectivity

- Pinged from PC1 (192.168.10.10) to Laptop1 (192.168.20.10):
>> ping 192.168.20.10
- Pinged from Laptop2 (192.168.20.11) to PC2 (192.168.10.11) to confirm communication.



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<lms TTL=128
Reply from 192.168.20.10: bytes=32 time<lms TTL=128
Reply from 192.168.20.10: bytes=32 time<lms TTL=128
Reply from 192.168.20.10: bytes=32 time<lms TTL=128

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
  
```

Conclusion

This lab demonstrated how to configure OSPF in a network with two routers and multiple devices. OSPF dynamically shared routing information between the routers, enabling seamless communication between devices in different networks. The shortest path algorithm ensured efficient routing across the topology.

LAB8: Configuring VLANs in a LAN Network

Introduction

In this lab, I configured Virtual LANs (VLANs) in Cisco Packet Tracer to segment a network into smaller logical networks. VLANs improve network performance and security by isolating traffic between groups of devices, even if they are physically connected to the same switch. This is particularly useful in large networks where dividing traffic helps reduce congestion and increase manageability.

Network Topology

- One switch (Cisco 2960)
- One router (Router 1841)
- Six end devices (4 PCs and 2 laptops)

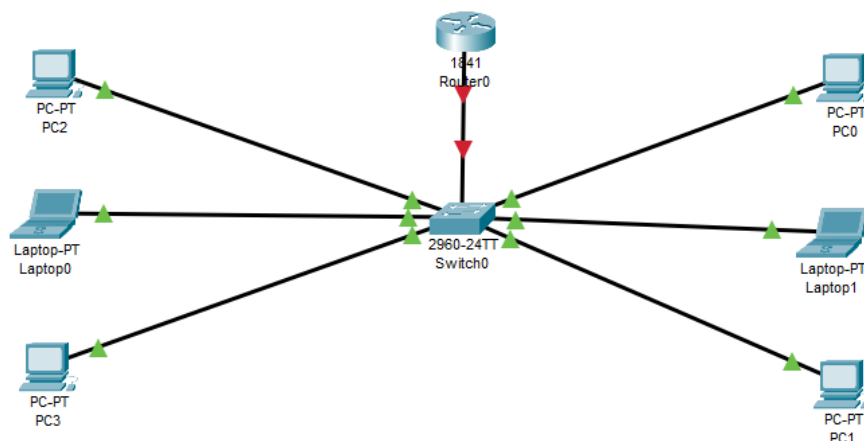
VLAN Plan

- **VLAN 10 (HR Department):** 192.168.10.0/24
 - Devices: PC0, PC1, Laptop0
- **VLAN 20 (IT Department):** 192.168.20.0/24
 - Devices: PC2, PC3, Laptop1

Practical Work

1. Physical Setup

- Connected PC1 and PC2 to the switch using straight-through cables.
- Connected Laptop1 and Laptop2 to the switch using straight-through cables.
- Connected the switch to the router's FastEthernet0/0 interface for inter-VLAN routing.



2. VLAN Configuration on the Switch

- Accessed the switch's CLI and created VLANs:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
```

Assigned ports to VLANs:

- **VLAN 10:**

```
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface FastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

- **VLAN 20:**

```
Switch(config)# interface FastEthernet0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface FastEthernet0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
```

- Configured a trunk link to the router for inter-VLAN routing:

```
Switch(config)# interface FastEthernet0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

3. Configuring the Router for Inter-VLAN Routing

- Accessed the router's CLI and enabled sub-interfaces for each VLAN:

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

```
Router(config)# interface FastEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

- Enabled the main interface:

```
Router(config)# interface FastEthernet0/0
Router(config-if)# no shutdown
```

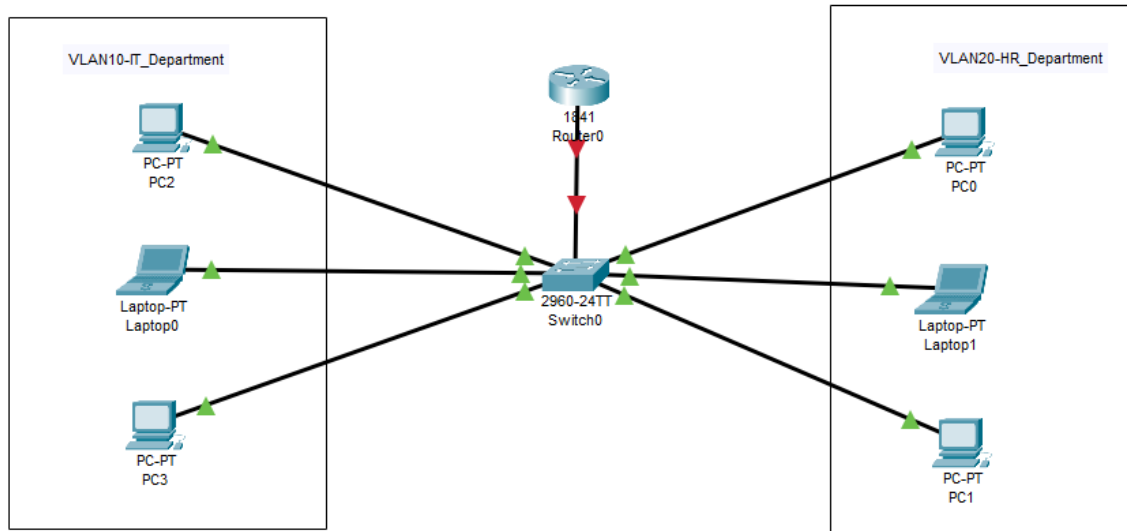
4. Configuring End Devices

- **PC1 and Laptop1 (VLAN 10):**
 - IP Address: 192.168.10.2, 192.168.10.3
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
- **PC2 and Laptop2 (VLAN 20):**
 - IP Address: 192.168.20.2, 192.168.20.3
 - Subnet Mask: 255.255.255.0

- Default Gateway: 192.168.20.1

5. Testing Connectivity

- Verified communication within the same VLAN by pinging devices in VLAN 10 and VLAN 20 separately:
`ping 192.168.10.3`
`ping 192.168.20.3`
- Tested inter-VLAN communication by pinging between devices in VLAN 10 and VLAN 20:
`ping 192.168.20.2`
`ping 192.168.10.2`



Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)# name IT
```

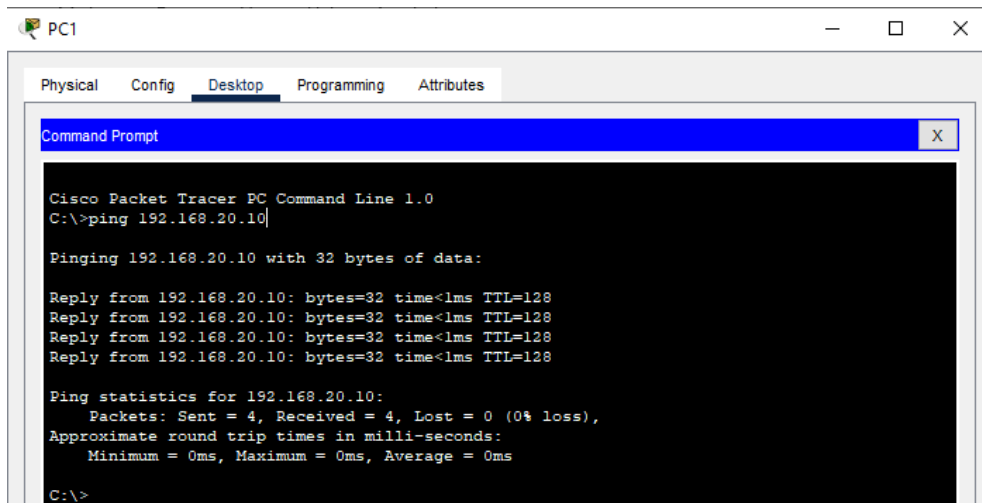
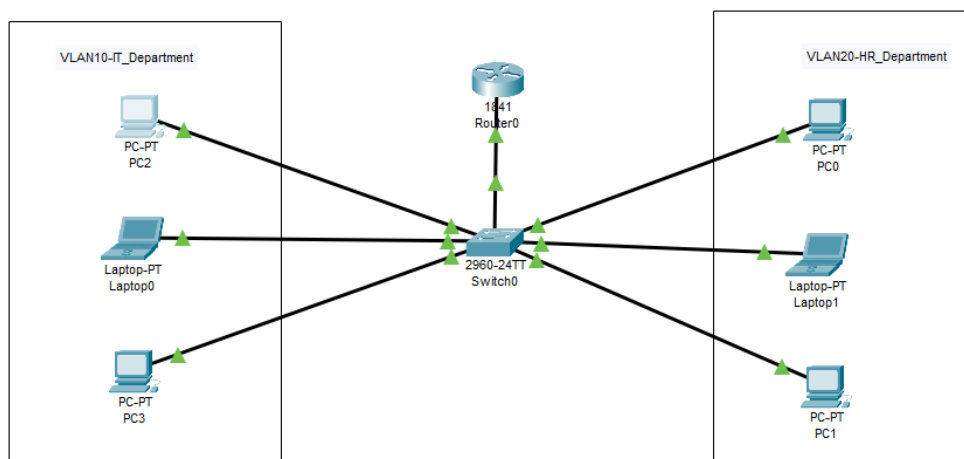
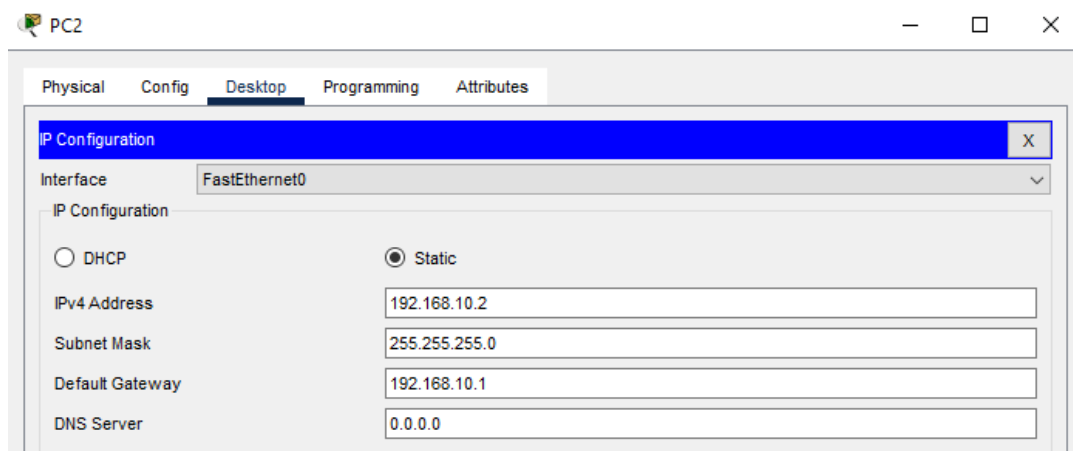
IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)# name IT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)# exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)# exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
```

IOS Command Line Interface

```
% Invalid input detected at '^' marker.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface FastEthernet0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
```



Conclusion

This lab demonstrated the setup of VLANs to segment a network into logical groups and configure inter-VLAN routing using a router. VLANs enhanced security and performance by isolating traffic, while inter-VLAN routing enabled communication between devices in different VLANs.

LAB9: Installing a Virtual Linux Machine and Exploring Networking Commands

Objective

1. To install a virtual Linux machine using virtualization software.
2. To explore and execute essential Linux networking commands for configuring and troubleshooting networks.

Apparatus

- A computer with internet access.
- Virtualization software: VirtualBox or VMware Workstation.
- ISO image of a Linux distribution (e.g., Ubuntu, Fedora, or Debian).

Part 1: Installing a Virtual Linux Machine

Step-by-Step Guide

1. **Download Required Software:**
 - Download VirtualBox or VMware Workstation from their official websites.
 - Obtain the ISO image of a Linux distribution (e.g., Ubuntu).
2. **Install Virtualization Software:**
 - Run the installer for VirtualBox or VMware.
 - Follow the installation prompts to complete the setup.
3. **Create a Virtual Machine:**
 - Open the virtualization software and click on "New" to create a new virtual machine.
 - Enter a name for your virtual machine (e.g., "Linux Networking VM").
 - Select the type as "Linux" and the version matching your ISO file.
4. **Allocate Resources:**
 - Assign memory (e.g., 4 GB RAM) and create a virtual hard disk (e.g., 20 GB).
5. **Attach the ISO File:**
 - Go to the VM's settings and attach the Linux ISO file under "Storage."
6. **Install Linux:**
 - Start the VM, boot from the ISO, and follow the Linux installation wizard.

- Complete the installation and reboot the virtual machine.

Part 2: Exploring Networking Commands

Once the Linux system is installed and running, execute the following networking commands to understand their functionality:

1. ifconfig / ip addr

- **Purpose:** Displays or configures network interfaces.
- **Usage:**

```
ifconfig
```

```
ip addr show
```

- **Example Output:** Displays IP address, subnet mask, and interface status.

2. ping

- **Purpose:** Checks connectivity to another device or server.
- **Usage:**

```
ping <hostname_or_IP>
```

- **Example:**

```
ping google.com
```

- **Output:** Displays round-trip time for packets sent to the destination.

3. netstat / ss

- **Purpose:** Displays network connections, routing tables, and statistics.
- **Usage:**

```
netstat -an
```

```
ss -tunlp
```

- **Example Output:** Lists active connections and listening ports.

4. traceroute

- **Purpose:** Displays the route packets take to a destination.
- **Usage:**

```
traceroute <hostname_or_IP>
```

- **Example:**

```
tracert google.com
```

- **Output:** Shows each hop in the route and its latency.

5. nslookup / dig

- **Purpose:** Queries DNS records to resolve domain names.
- **Usage:**

```
nslookup <hostname>
```

```
dig <hostname>
```

- **Example:**

```
nslookup google.com
```

```
dig google.com
```

- **Output:** Displays DNS server response, including IP addresses.

6. curl / wget

- **Purpose:** Fetches data from a URL or downloads files.
- **Usage:**

```
curl <URL>
```

```
wget <URL>
```

- **Example:**

```
curl http://example.com
```

```
wget http://example.com/file.txt
```

- **Output:** Downloads or displays the content of the URL.

7. route

- **Purpose:** Displays or modifies the routing table.
- **Usage:**

```
route -n
```

- **Output:** Shows routing table entries, including gateways and network interfaces.

8. arp

- **Purpose:** Displays and manipulates the ARP table (used for IP-to-MAC address mapping).
- **Usage:**

```
arp -a
```

- **Output:** Lists IP and MAC addresses of connected devices.

9. iptables

- **Purpose:** Configures the firewall rules for packet filtering.
- **Usage:**

```
sudo iptables -L
```

- **Output:** Displays the current firewall rules.

10. nmcli

- **Purpose:** Manages network connections using NetworkManager.
- **Usage:**

```
nmcli device show
```

```
nmcli connection up <connection_name>
```

- **Example:**

```
nmcli connection show
```

- **Output:** Displays active and available network connections.

Conclusion

By completing this lab, I have:

- Successfully installed and configured a virtual Linux machine.
- Learned to execute essential networking commands for diagnosing and troubleshooting networks in a Linux environment.